SHADOWX

ShadowX User Manual

ICT3215 Digital Forensics PA-G 35	
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Table of Content

Introduction

Welcome to ShadowX, A Monero Analysis web tool

Currently, ShadowX is developed under the Monero Stagenet environment and in offline mode. The Monero stagenet service will be hosted at the device port 38081.

Installation and Setup

As this tool is currently only developed for usage in Kali Linux and suitable only in Linux environment, please take note of the following steps apply only in this environment

To install this tool, follow this step:

1. Clone the repository

```
git clone https://github.com/echoezz/ShadowX.git
```

2. Place the project directory ShadowX at your home directory of your Kali VM (Eg. /home/kali/ShadowX)

3. As github does not publish empty directories, you will need to create 3 folders inside the ShadowX folder. (Follow the naming convention)

   - Firstly, create the folder `uploadedFiles`
   - **Then inside the folder `uploadedFiles`, create the folder `stagenet`**
   - **Lastly, inside the folder `stagenet`, create the folder `lmdb`**

   Your folder structure should look like this:

```
ShadowX/
├── templates/
├── uploadedFiles/
│   ├── stagenet
│   │   ├── lmdb
├── static/
├── analysis/
├── monero-x86_64-linux-gnu-v0.18.4.4/
├── app.py
├── img/
```

4. Navigate to the folder monero-x86_64-linux-gnu-v0.18.4.4 and execute the command to give monerod executable permission.

```
chmod +x monerod
```

5. To start the web application, navigate to the ShadowX directory folder and run the following command in the terminal

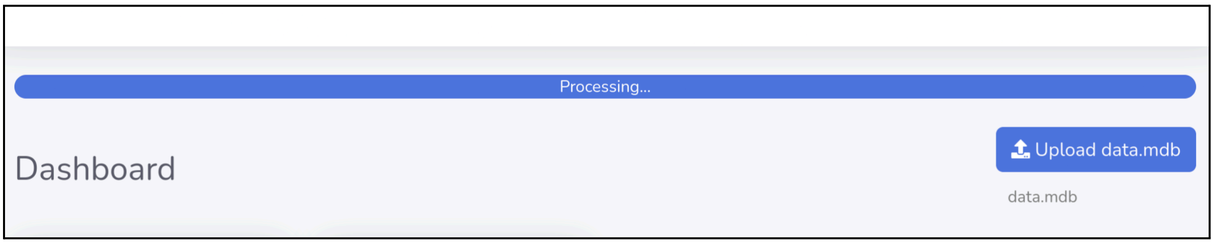
```
python3 app.py
```

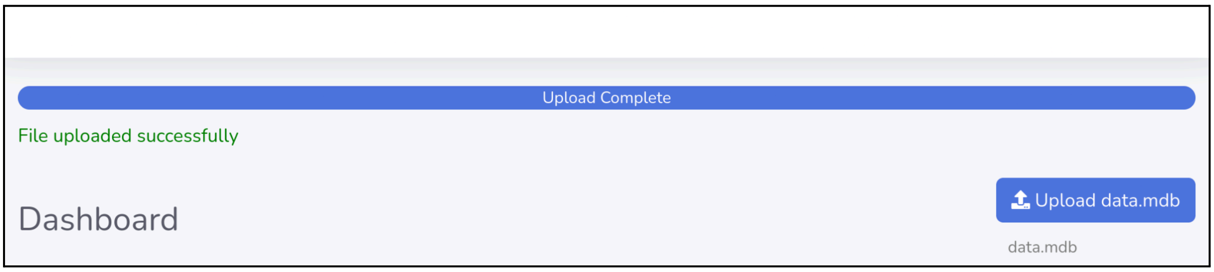
Open your browser and visit the page at localhost:5000 to start using ShadowX

Using ShadowX

Uploading of data.mdb file

At the **index** page, click the button “Upload data.mdb” button to upload the data.mdb file you have and wait for the file to finish uploading





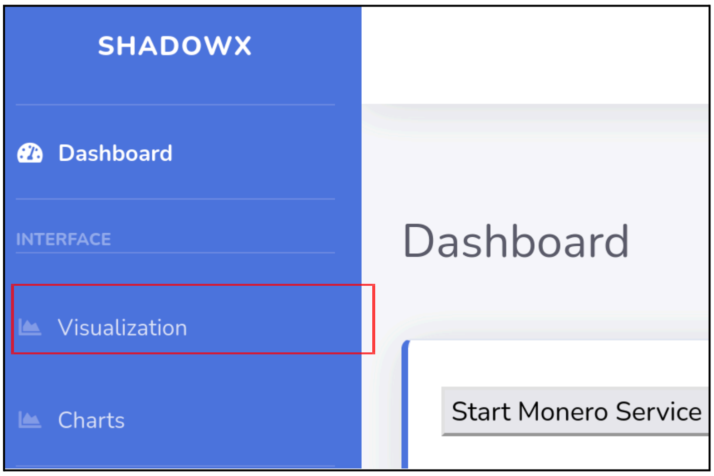
Starting the Monero service

At the index page, after uploading the data.mdb file successfully, start the Monero service by clicking the button “Start Monero Service”

Once started, we can begin to perform the analysis

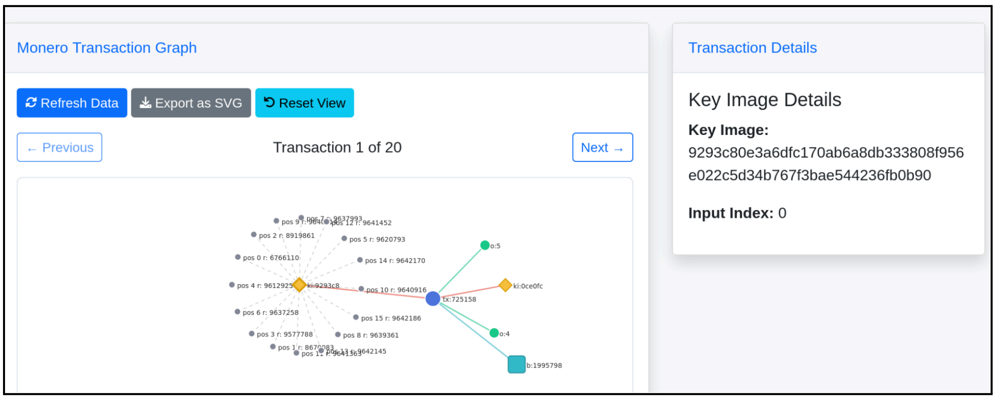
Visualization

Users can navigate to the visualization page by pressing on the visualization tab at the side. At the visualization page, once the Monero service is started, users can view the different transactions within the data.mdb file
(For this project, we only scale up to 20 transactions to reduce overhead)



Users can interact with the nodes by clicking on it, which will return the respective information as such:

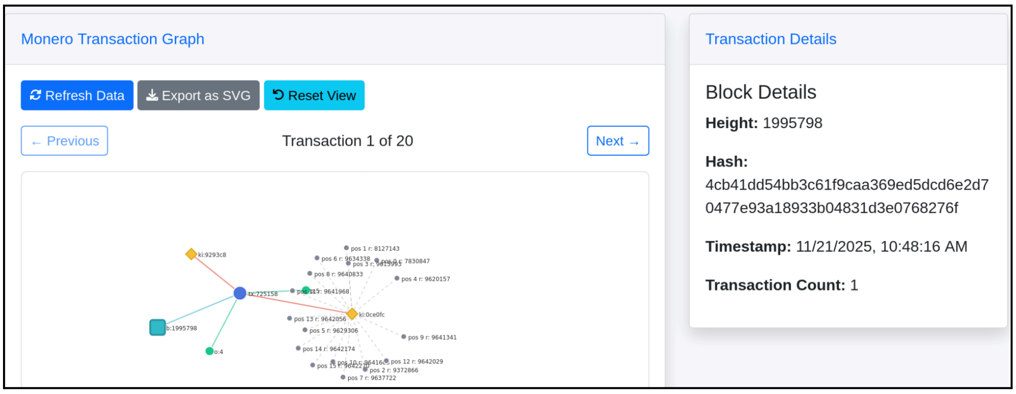
- Yellow nodes for Key Image Details



- Blue nodes for Transaction details

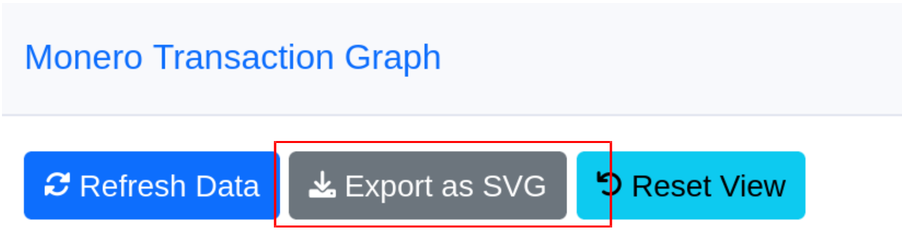
- Green node for Output details

- Square node (turquoise) for Block Details



- Grey node for Ring Member Details

Users can also choose to export the visualization if they wish to do so via the “Export as SVG” button

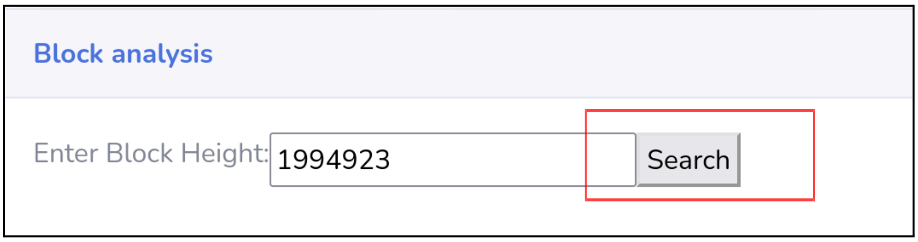


Block analysis lookup

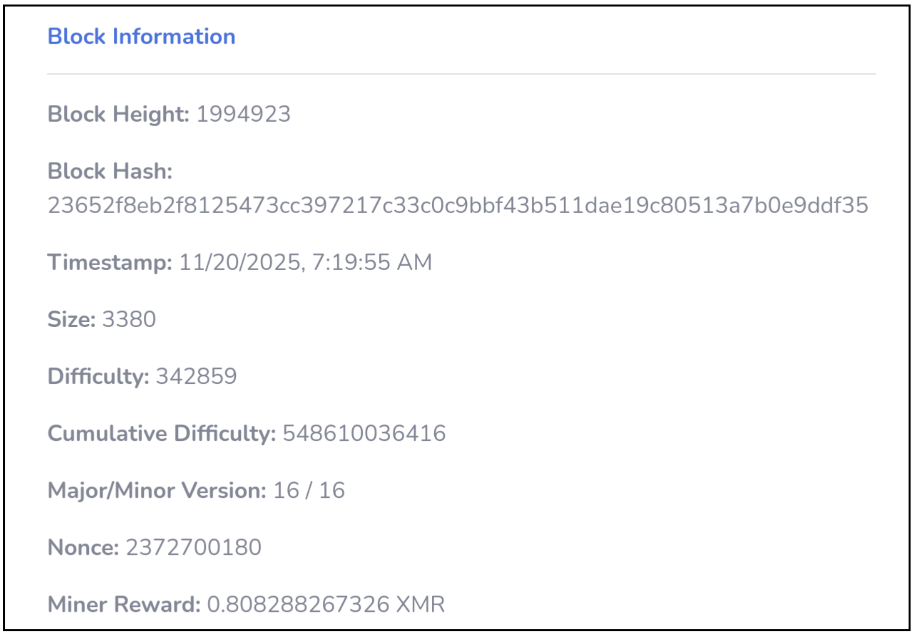
If users require more information, they can perform block analysis by looking up values they see from the visualization, after interacting with the nodes.

Get block by height segment

Users can input a block height of their interest and perform a search

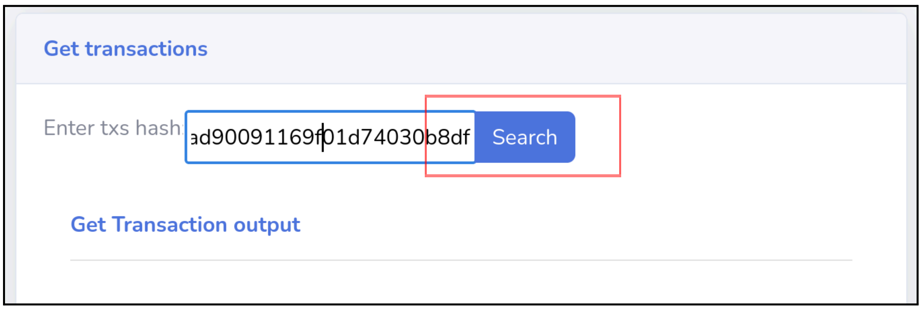


They will then be able to gain more information from the block height as follows for analysis purposes:

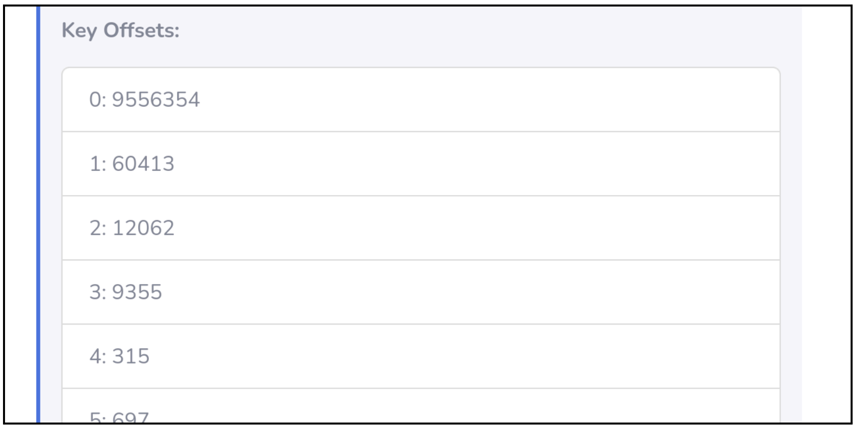


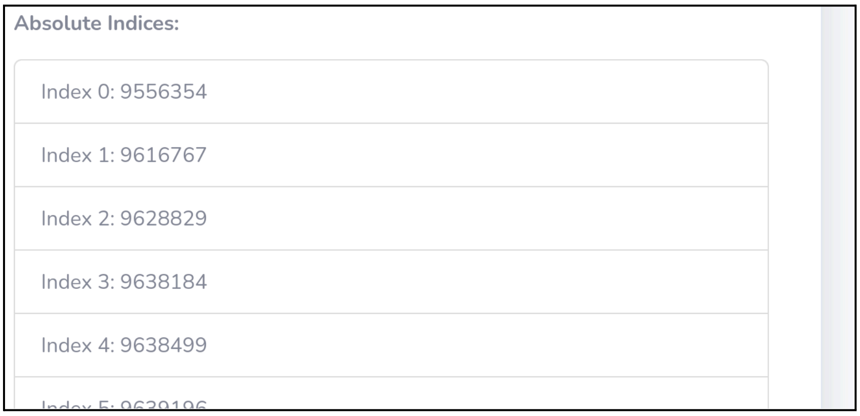
Get transaction segment

Users can also perform a get_transaction lookup by supplying the transactional hash they retrieved from the visualization page earlier after their interaction with the nodes



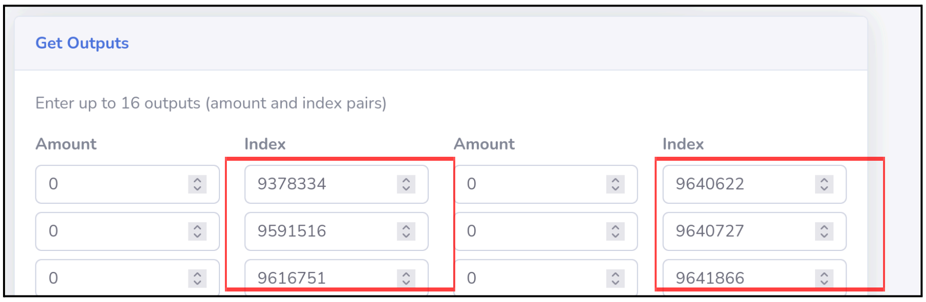
They will then be able to retrieve information such as the key image, key offsets and absolute indices

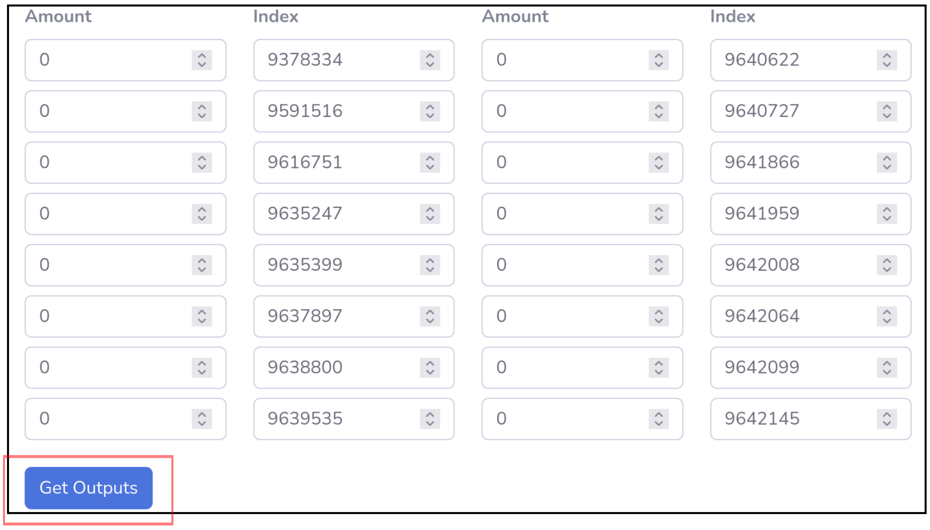




Get outs segment

If they wish to make use of the get_outs function, users can take the 16 absolute indices and fill up accordingly to the index position:

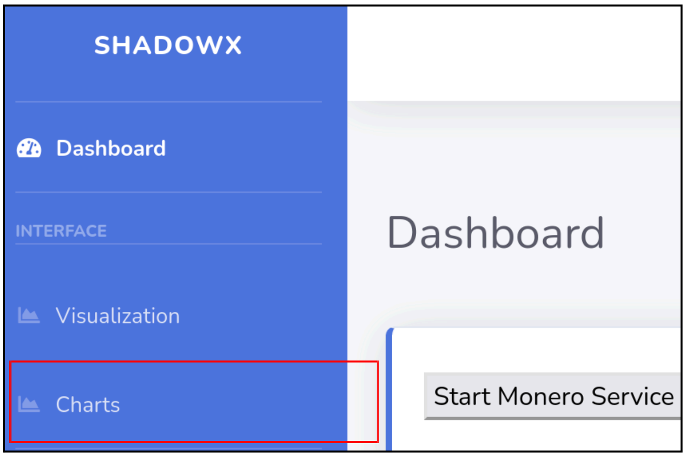




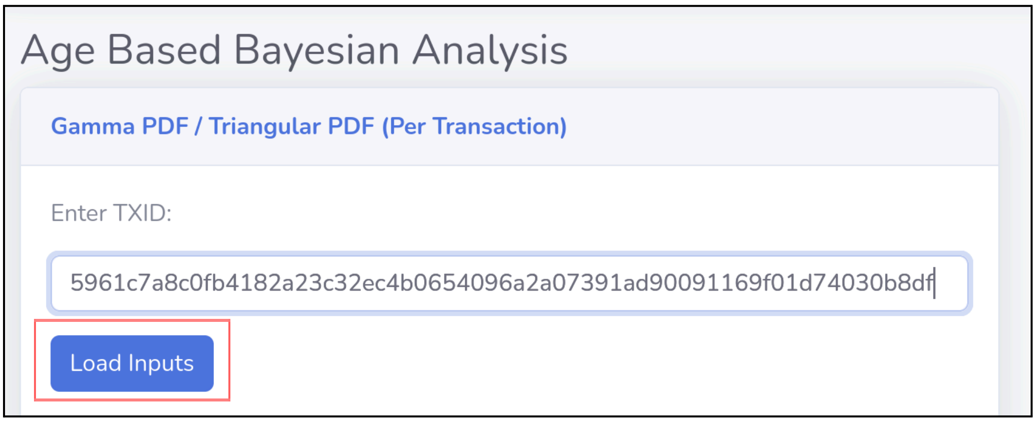
Once they submit via the “Get Outputs” button, they can retrieve information such as Block height, Key, Mask, Transaction ID:

Age Based Bayesian Analysis

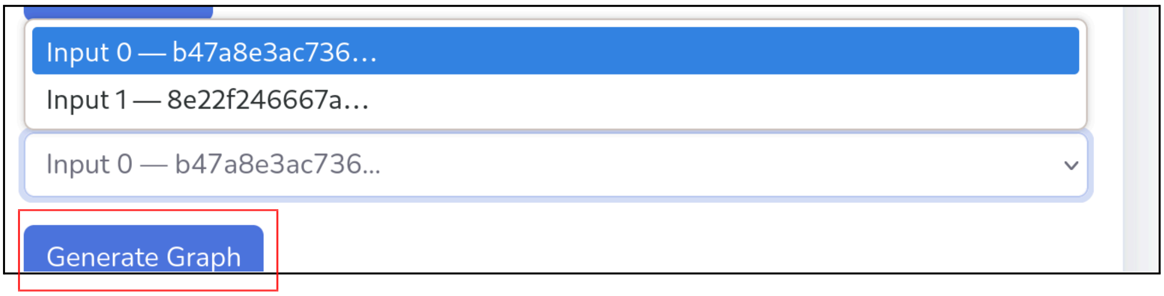
If the user wished to perform Age Based Bayesian Analysis on attempt to analyse the decoys previously seen from the visualization page to determine the likely real spent of a transaction, they can head over to the charts page with a transactional hash taken from the visualization page



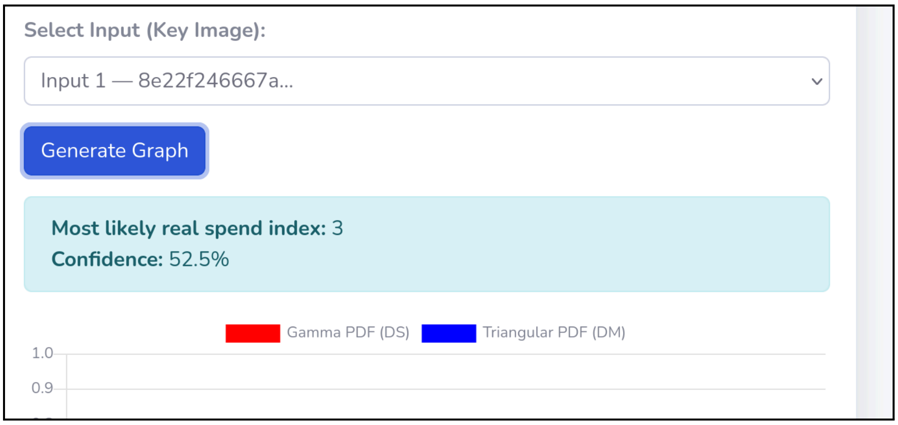
They can perform a search with the transactional hash and click on the "Load Inputs" button

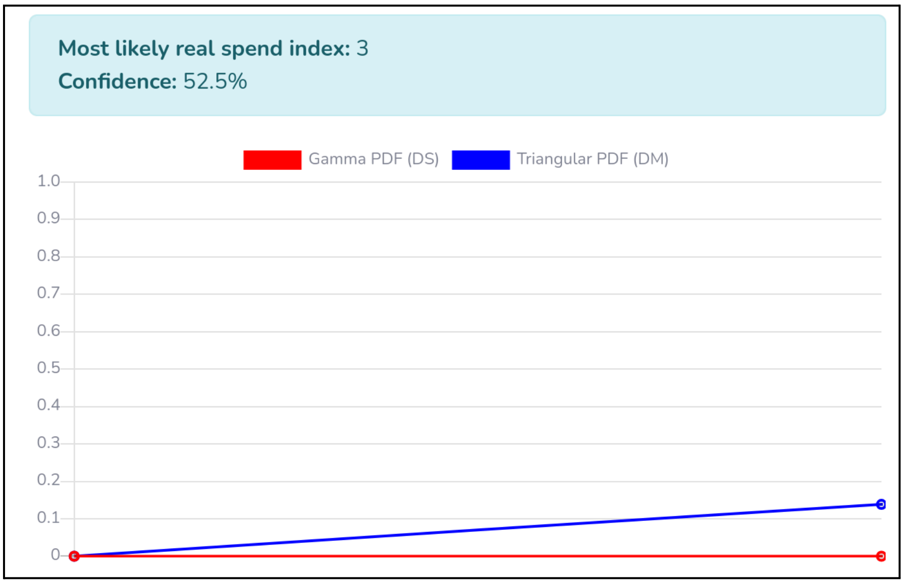


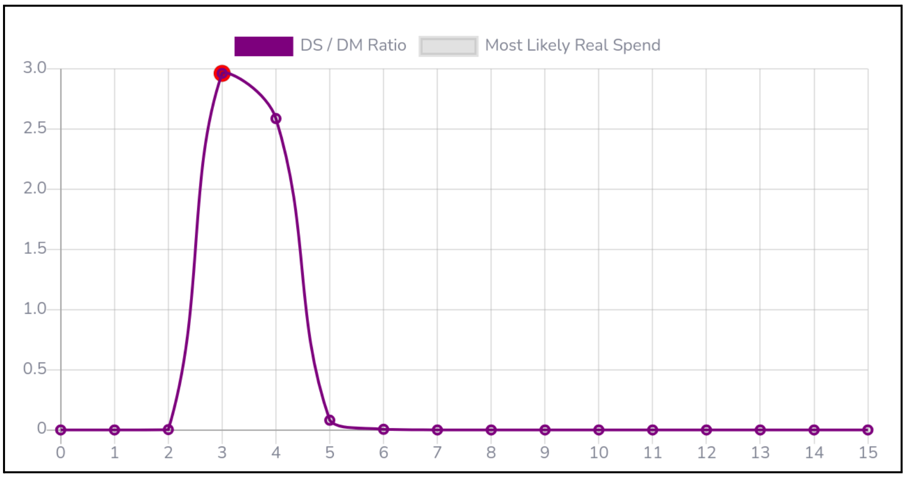
If the transactional hash has multiple key images, it will show up accordingly for the user to choose from. They can then generate a graph by clicking on the "Generate Graph" button



After clicking on the "Generate Graph" button, they can view the following analysis that makes use of Age Based Bayesian Analysis to try and locate the likely real spent of a transaction.


Select Input (Key Image):
Input 1 — 8e22f246667a...
Generate Graph
Most likely real spend index: 3
Confidence: 52.5%
Gamma PDF (DS)
Triangular PDF (DM)
1.0
0.9

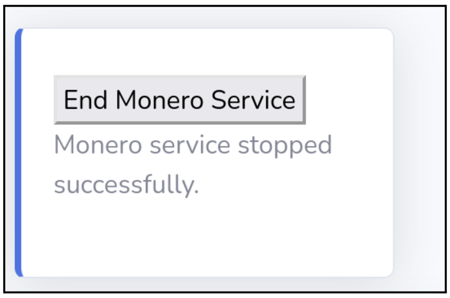

Most likely real spend index: 3
Confidence: 52.5%
Gamma PDF (DS)
Triangular PDF (DM)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0


DS / DM Ratio
Most Likely Real Spend
3.0
2.5
2.0
1.5
1.0
0.5
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Terminating ShadowX

Terminating Monero Service

Once a user is done using ShadowX, they can proceed to the index page and click on the "End Monero Service" button to terminate ShadowX and the Monero service properly.



Users can also run the following command on their terminal in Kali Linux to ensure the Monero service is shutdown successfully.

```
ps aux | grep monerod
```

Terminating ShadowX

To terminate ShadowX, at the terminal where app.py was launched, execute Ctrl C to end the application.

End of user manual